

A scalar reduction of the two-row $d=4$ fiber-vanishing law

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Abstract

For $\lambda = (a, b)$ with $a + b = 2m$, set $G_{(a,b)}(i) = \langle s_{(a,b)}, \psi^m \rangle$, where $\psi = h_2 + i e_2$ and $i = \sqrt{-1}$. The two-row $d=4$ fiber-vanishing conjecture asserts

$$G_{(a,b)}(i) = 0 \iff (a, b) = (2, 2).$$

This note reduces the conjecture to an *elementary one-dimensional scalar positivity statement* and pins the residual gap precisely. Through a recessive continued-fraction representation we pass to coordinates (p_n, q_n) in which the law reads $q_n > 0$, and we exhibit a p -free scalar comparison sequence Q_n that would force $q_n > 0$ via two clean lemmas: a domination $q_n \geq Q_n$ and a scalar positivity $Q_n > 0$. The sharp statement is $q_n \geq 0$ for all $m \geq 3$ with equality *iff* $(m, n) = (3, 1)$, which is exactly the $(2, 2)$ fiber. Neither comparison lemma yet has a closed-form proof; we localise the entire binding margin to a single scalar step $n = 2 \rightarrow 1$ around a cubic root and state the concrete remaining target.

1 Setup and the continued fraction

Write $\lambda = (a, b)$ with $a \geq b \geq 0$ and $a + b = 2m$. Define

$$r_l = [u^l] (u^2 + (1 + i)u + 1)^m, \quad 0 \leq l \leq 2m,$$

so that r_l is palindromic, $r_l = r_{2m-l}$ (the generating polynomial is self-reciprocal). The fiber value is the adjacent difference

$$G_{(a,b)}(i) = r_b - r_{b-1}. \quad (1)$$

Equivalently, writing $\phi_j := r_{m-j}$, the ϕ_j are the Chebyshev/Fourier coefficients of a *pure power*: with $c = \cos \theta$,

$$(1 + i + 2 \cos \theta)^m = (2c + 1 + i)^m = 2^m (c - c_0)^m, \quad c_0 = -\frac{1+i}{2},$$

and $\phi_j = \frac{1}{2\pi} \int_0^{2\pi} (1 + i + 2 \cos \theta)^m e^{ij\theta} d\theta$. This is the form recorded in the companion note of 2026-06-05; here we exploit it to obtain a terminating continued fraction.

Because $(u^2 + (1 + i)u + 1)^m$ satisfies a three-term contiguous relation in its coefficients, the ratios $\xi_j := (m + j + 1) \phi_{j+1} / \phi_j$ obey

$$\xi_m = 0, \quad \xi_{n-1} = \frac{A_n}{(1 + i)n + \xi_n}, \quad A_n = (m + n)(m - n + 1), \quad (2)$$

the recessive (terminating) branch of the underlying recurrence; this is the unique solution that stays finite, equivalently the Pincherle minimal solution. The law (1) translates into a sign condition on the tails. Setting

$$y_n := \frac{\xi_n}{(1 + i)n}, \quad P_n := 1 + \operatorname{Re} y_n + \operatorname{Im} y_n,$$

one finds, after clearing the $(1+i)$ phase, that

$$G_{(a,b)}(i) \neq 0 \quad \text{for the relevant } (a,b) \iff P_n > 0, \quad (3)$$

with the boundary index $n = b$ controlling the shape (a,b) . The two-row law is thus the single positivity statement $P_n > 0$ for all n in the relevant range, with the conjectured unique failure at the $(2,2)$ fiber.

2 The scalar reduction

The content of this note is to strip (2) down to a genuinely one-dimensional comparison. Introduce the affine coordinates

$$p_n := \operatorname{Re} \xi_n - (m - n), \quad q_n := \operatorname{Im} \xi_n + n. \quad (4)$$

The key dictionary entry is

$$q_n = n P_n = n(1 + \operatorname{Re} y_n + \operatorname{Im} y_n),$$

so that

$$\boxed{\text{the law } \iff q_n > 0} \quad (5)$$

for the relevant n , with the binding case $n = 1$.

Proposition 1 ((p, q) recursion). *With $D_n := q_n^2 + (m + p_n)^2$, the continued fraction (2) is equivalent to*

$$q_{n-1} = (n-1) - q_n \frac{A_n}{D_n}, \quad (6)$$

$$p_{n-1} = (m-n+1) \frac{mn - p_n(m-n+p_n) - q_n^2}{D_n}, \quad (7)$$

with seeds (exact)

$$\xi_{m-1} = 1 - i \implies p_{m-1} = 0, \quad q_{m-1} = m - 2, \quad p_m = 0, \quad q_m = m.$$

Proof. Substitute $\xi_n = (m-n) + p_n + i(-n + q_n)$ into (2), rationalise the denominator $(1+i)n + \xi_n = (m + p_n) + i(q_n)$ (its modulus squared is exactly D_n), and separate real and imaginary parts. The seed $\xi_{m-1} = A_m / ((1+i)m)$ with $A_m = 2m \cdot 1 = 2m$ gives $\xi_{m-1} = 2m / ((1+i)m) = 1 - i$, hence $p_{m-1} = \operatorname{Re}(1 - i) - 1 = 0$ and $q_{m-1} = \operatorname{Im}(1 - i) + (m - 1) = m - 2$. $\square$

The decisive observation is that the q -recursion (6) couples to p *only* through the denominator $D_n = q_n^2 + (m + p_n)^2$, and there only through $(m + p_n)^2$. If $p_n \geq 0$ then $D_n \geq q_n^2 + m^2$, so replacing p_n by 0 *decreases* the denominator and (since $q_n > 0$, $A_n > 0$) decreases the subtracted term, hence cannot lower q_{n-1} . This motivates the p -free *scalar comparison sequence*

$$Q_m = m, \quad Q_{n-1} = (n-1) - \frac{Q_n A_n}{Q_n^2 + m^2}. \quad (8)$$

Proposition 2 (Reduction). *The two-row $d=4$ law (in the form $q_n > 0$, (5)) follows from the two statements*

(F1) nonnegativity of p : $p_n \geq 0$ for all n ;

(F3) domination: $q_n \geq Q_n$ for all n ;

(F4) scalar positivity: $Q_n > 0$ for all $m \geq 4$ and all n , while at $m = 3$ one has $Q_1 = 0$ exactly.

Indeed for $m \geq 4$, $q_n \geq Q_n > 0$ gives the strict law; for $m = 3$, $q_1 = 0$ exactly yields the unique fiber vanishing.

Proof. Assume (F1). Then $D_n \geq q_n^2 + m^2$. Given (F3)+(F4): for $m \geq 4$, $q_n \geq Q_n > 0$. For $m = 3$ the comparison sequence satisfies $Q_1 = 0$ exactly, and the recursion (6) with the actual $p_1 = 0$ forces $q_1 = 0$ as well, i.e. $P_1 = 0$, i.e. $G_{(2,2)}(i) = 0$ by (3). $\square$

Thus (F1), (F3), (F4) are all genuinely scalar: (F4) involves the single recursion (8) with no complex arithmetic at all, and (F1)/(F3) compare two real scalar sequences. The full two-variable complex dynamical system of (2) has been reduced to comparison against one explicit real recursion.

3 The sharp theorem

The strict form “ $P_n > 0$ for all n ” is *false* at exactly one place, and the correct statement is the following.

Theorem 1 (Sharp form). *For all $m \geq 3$ and all relevant n ,*

$$P_n \geq 0, \quad \text{equivalently} \quad q_n \geq 0,$$

with equality iff $(m, n) = (3, 1)$. This single boundary vanishing is exactly the $(2, 2)$ fiber, $G_{(2,2)}(i) = 0$.

This is precisely the conjectured biconditional: $G_{(a,b)}(i) = 0 \iff (a, b) = (2, 2)$. The shape $(2, 2)$ corresponds to $m = 2$ in $a + b = 2m$; the boundary index in the ξ -recursion that realises the vanishing is $(m, n) = (3, 1)$ in the normalisation above, the unique zero of P_n . The asymptotic margin is sharp: $q_1 = P_1 \rightarrow \frac{1}{4}$ as $m \rightarrow \infty$, so the positivity is never comfortable — the sequence approaches 0 from above and only just clears it.

Remark 1. The boundary at $n = 1$ explains why every prior “window” or “box-envelope” attempt failed: the binding constraint is not a fixed band but the single endpoint step, and the available slack there shrinks to a fixed positive limit $1/4$ rather than growing. There is no room to be lossy.

4 The residual gap (honest accounting)

Neither (F3) nor (F4) has a closed-form proof yet. We record precisely where the elementary arguments break, so the next attempt does not repeat them.

The slow manifold is a cubic. Writing $t = n/m$ and tracking the leading behaviour of Q_n/n as $m \rightarrow \infty$, the comparison recursion (8) has a slow manifold $\varphi = \varphi(t)$ satisfying the cubic

$$(\varphi - t)(\varphi^2 + 1) = -\varphi(1 - t^2). \tag{9}$$

Its relevant root is a Cardano radical with no elementary envelope; in particular there is no rational or quadratic barrier L_n trapping Q_n from below uniformly in m . This is why (F4), though numerically clean, resists a one-line induction.

The map is non-monotone. Fix n and view $g(q) = (n-1) - q A_n/(q^2 + m^2)$ as a function of q . Then g is *decreasing* for $q < m$ (the regime in play, since $q_n \leq n \leq m$) and increasing for $q > m$. Hence the naive monotone-iteration arguments do not apply: a lower bound on q_n does not directly propagate to a lower bound on q_{n-1} .

One-sided bounds fail. Optimising g over q one-sidedly is too lossy: the unconstrained $\max_q g$ at $n = 2$ already gives

$$\max_q \left[1 - \frac{q A_2}{q^2 + m^2} \right] = 1 - \frac{A_2}{2m} = -\frac{(m-2)^2}{2m} < 0,$$

so any argument that does not use the actual location of q_2 (not just a one-sided bound on it) is doomed. The positivity is genuinely two-sided.

Where the margin lives. The entire binding margin concentrates in the single scalar step $n = 2 \rightarrow 1$. What is needed is a *two-sided envelope*

$$0.6 \lesssim Q_2 \lesssim 0.75$$

of accuracy $O(1/m)$ around the relevant root of the cubic (9). Numerically Q_2 rises monotonically from ≈ 0.719 ($m = 10$) toward 0.75, and Q_1 from this band lands at $Q_1 \rightarrow \frac{1}{4}^+$. A rigorous proof of (F4) reduces to trapping Q_2 in such a band with error $o(1)$ uniformly in m — a finite, concrete analytic target. The coupled pair (F1)+(F3) is the twin obstruction: the margin is asymptotically sharp, so p and q cannot be decoupled by a slack bound; one needs a joint lower envelope tying them together.

Concrete next target. Prove the two-sided estimate

$$Q_2(m) = \frac{3}{4} - \frac{c}{m} + o(1/m) \quad (c > 0),$$

equivalently a uniform $O(1/m)$ approximation of Q_n by the cubic root $\varphi(n/m)$ for $n = O(1)$, and separately establish (F1)+(F3) jointly via a single invariant region for the pair (p_n, q_n) rather than for each coordinate in isolation.

5 Computational verification

All identities and the four structural facts were verified with the script `proof_skeleton.py` (with helper `seqs.py`) in `~/projects/scratch/2026-06-06-forkI/`, using `mpmath` at 40 decimal digits. The two constructions of ξ — directly from the coefficient ratios $(m+j+1)\phi_{j+1}/\phi_j$ and from the terminating continued fraction (2) — agree to machine precision across $m \in \{5, 10, 20, 50, 100\}$. The exhaustive check over $m = 3, \dots, 400$ (and additionally $m = 419$) reports: $\min_n p_n = 0$ (F1, attained only at the seeds); $\min_n (q_n - Q_n) = 0$ (F3, with equality only at $n \in \{m, m-1, m-2\}$, strict below); $\min_{m \geq 4, n} Q_n \approx 0.184$ at $m = 5$ (F4); and $\min_n P_n = 0$ attained *only* at $(m, n) = (3, 1)$, confirming Theorem 1. The binding margin Q_1 is observed to rise to $1/4$ ($Q_1 \approx 0.2494$ at $m = 400$; $\min_n P_n \approx 0.2504$ at $m = 419$), matching the analytic limit.